

Main XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="15dp"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/etRno"
        android:layout_width="match_parent"
        android:layout_height="62dp"
        android:hint="Enter Roll Number"
        android:inputType="number" />

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="57dp"
        android:hint="Enter Student Name"
        android:inputType="text" />

    <RadioGroup
        android:id="@+id/radioGroup"
        android:layout_width="383dp"
        android:layout_height="45dp"
        android:orientation="horizontal">

        <RadioButton
            android:id="@+id/rbMale"
            android:layout_width="191dp"
            android:layout_height="60dp"
            android:layout_weight="2"
            android:checked="true"
            android:text="Male" />

        <RadioButton
            android:id="@+id/rbFemale"
            android:layout_width="172dp"
            android:layout_height="60dp"
            android:layout_weight="2"
            android:text="Female" />

    </RadioGroup>
```

```

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="62dp"
    android:orientation="horizontal">

    <EditText
        android:id="@+id/etDate"
        android:layout_width="312dp"
        android:layout_height="62dp"
        android:layout_gravity="center"
        android:layout_weight="1"
        android:editable="false"
        android:hint="Click image to Set Date" />

    <ImageButton
        android:id="@+id/imgDate"
        android:layout_width="70dp"
        android:layout_height="70dp"
        android:src="@drawable/baseline_date_range_24"
        android:autofillHints="date"
        android:contentDescription="date"

    />
</LinearLayout>
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="1dp"
    android:orientation="horizontal"
    >

    <Spinner
        android:id="@+id/spinner_cities"
        android:layout_width="match_parent"
        android:layout_height="62dp" />

</LinearLayout>

<LinearLayout
    android:layout_width="384dp"
    android:layout_height="wrap_content"
    android:orientation="horizontal">

    <Button
        android:id="@+id/course"
        android:layout_width="125dp"

```

```
    android:layout_height="wrap_content"
    android:text="Course">
```

```
</Button>
```

```
<TextView
    android:id="@+id/coursetv"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="course">
```

```
</TextView>
```

```
<Button
    android:id="@+id/time"
    android:layout_width="135dp"
    android:layout_height="match_parent"
    android:text="time">
```

```
</Button>
```

```
<TextView
    android:id="@+id/timetv"
    android:layout_width="75dp"
    android:layout_height="wrap_content"
    android:text="time">
```

```
</TextView>
```

```
</LinearLayout>
```

```
<LinearLayout
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal">
```

```
<CheckBox
    android:id="@+id/cricket"
    android:layout_width="107dp"
    android:layout_height="wrap_content"
    android:text="cricket">
```

```
</CheckBox>
```

```
<CheckBox
    android:id="@+id/singing"
    android:layout_width="115dp"
```

```
    android:layout_height="wrap_content"
    android:text="singing">
```

```
</CheckBox>
```

```
<CheckBox
    android:id="@+id/dancing"
    android:layout_width="112dp"
    android:layout_height="wrap_content"
    android:text="dancing">
```

```
</CheckBox>
```

```
</LinearLayout>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">
    <Button
        android:id="@+id/btnInsert"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Insert" />
    <Button
        android:id="@+id/btnUpdate"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Update" />
    <Button
        android:id="@+id/btnDelete"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Delete" />
    <Button
        android:id="@+id/btnsearch"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Search" />
```

```
</LinearLayout>
```

```
<ListView
    android:id="@+id/listView1"
    android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content" />

</LinearLayout>
```

MainActivity

```
package com.example.crud;

import android.app.DatePickerDialog;
import android.app.TimePickerDialog;
import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.DatePicker;
import android.widget.EditText;
import android.widget.ImageButton;
import android.widget.ListView;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.SimpleCursorAdapter;
import android.widget.Spinner;
import android.widget.TextView;
import android.widget.TimePicker;
import android.widget.Toast;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;
import java.util.Calendar;

public class MainActivity extends AppCompatActivity {
    EditText rollNo, name, date;
    DbHelper db;
    ImageButton imageButton;
    RadioGroup radioGroup;
    String radio = "Male";
    Button btnInsert, btnUpdate1, btnDelete, btnSearch;
    ListView listView;
    Button course, time;
    CheckBox cricket, singing, dancing;
    TextView timetv, coursetv;
```

```
String selectedCity,selectedCourse,selectedTime;
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {
```

```
    super.onCreate(savedInstanceState);
```

```
    setContentView(R.layout.activity_main);
```

```
    rollNo = findViewById(R.id.etRno);
```

```
    name = findViewById(R.id.etName);
```

```
    date = findViewById(R.id.etDate);
```

```
    imageButton = findViewById(R.id.imgDate);
```

```
    radioGroup = findViewById(R.id.radioGroup);
```

```
    btnInsert = findViewById(R.id.btnInsert);
```

```
    btnUpdate1 = findViewById(R.id.btnUpdate);
```

```
    btnDelete = findViewById(R.id.btnDelete);
```

```
    btnsearch=findViewById(R.id.btnsearch);
```

```
    listView = findViewById(R.id.listView1);
```

```
    course=findViewById(R.id.course);
```

```
    time=findViewById(R.id.time);
```

```
    cricket=findViewById(R.id.cricket);
```

```
    singing=findViewById(R.id.singing);
```

```
    dancing=findViewById(R.id.dancing);
```

```
    timetv=findViewById(R.id.timetv);
```

```
    coursetv=findViewById(R.id.coursetv);
```

```
    db = new DbHelper(this);
```

```
//showdatainlistview() calling
```

```
    showDataInListView1();
```

```
radioGroup.setOnCheckedChangeListener(new RadioGroup.OnCheckedChangeListener() {
```

```
    @Override
```

```
    public void onCheckedChanged(RadioGroup group, int checkedId) {
```

```
        RadioButton id = findViewById(checkedId);
```

```
        radio = id.getText().toString();
```

```
        Toast.makeText(MainActivity.this, "" + radio, Toast.LENGTH_SHORT).show();
```

```
    }
```

```
});
```

```
imageButton.setOnClickListener(new View.OnClickListener() {
```

```
    @Override
```

```
    public void onClick(View v) {
```

```
        Calendar cal = Calendar.getInstance();
```

```
        int year1 = cal.get(Calendar.YEAR);
```

```
        int month1 = cal.get(Calendar.MONTH);
```

```
        int date1 = cal.get(Calendar.DATE);
```

```
        DatePickerDialog datePickerDialog = new DatePickerDialog(MainActivity.this, new
```

```

        DatePickerDialog.OnDateSetListener() {
            @Override
            public void onDateSet(DatePicker view, int year, int month, int dayOfMonth) {
                String d = dayOfMonth + "/" + (month + 1) + "/" + year;
                date.setText(d);
            }
        }, year1, month1, date1);
        datePickerDialog.show();
    }
});

```

```

time.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        timedia();
    }
});

```

```

Spinner spinner = findViewById(R.id.spinner_cities);
String[] cities = {"Select City", "New York", "Los Angeles", "Chicago", "Houston", "Miami"};
ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
android.R.layout.simple_spinner_item, cities);
adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
spinner.setAdapter(adapter);
spinner.setOnItemClickListener(new AdapterView.OnItemClickListener() {
    @Override
    public void onItemClick(AdapterView<?> parent, View view, int position, long id) {
        String selectedCity = parent.getItemAtPosition(position).toString();
        if (position != 0) {
            Toast.makeText(MainActivity.this, "Selected: " + selectedCity,
Toast.LENGTH_SHORT).show();
        }
    }
    @Override
    public void onNothingSelected(AdapterView<?> parent) {

    }
});

```

```

course.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        selectcourse();
    }
}

```

```
});
```

```
ArrayList<String> selectedCheckboxes = new ArrayList<>();  
if (cricket.isChecked())  
    selectedCheckboxes.add(cricket.getText().toString());  
if (singing.isChecked())  
    selectedCheckboxes.add(singing.getText().toString());  
if (dancing.isChecked())  
    selectedCheckboxes.add(dancing.getText().toString());  
String hobby = selectedCheckboxes.toString();
```

```
//Crud operation
```

```
btnInsert.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        int r = Integer.parseInt(rollNo.getText().toString());  
        String n = name.getText().toString();  
        String d = date.getText().toString();  
  
        //String t= timetv.getText().toString();  
  
        db.insert(r, n, radio, d,selectedTime,selectedCity,selectedCourse,hobby);  
  
        showDataInListView1();//after inser data chang in list  
    }  
});
```

```
btnUpdate1.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        int r = Integer.parseInt(rollNo.getText().toString());  
        String n = name.getText().toString();  
        String d = date.getText().toString();  
        db.update(r,n, radio, d,selectedTime,selectedCity,selectedCourse,hobby);  
  
        showDataInListView1();//after update data chang in list  
    }  
});
```

```
btndelete.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        int roll = Integer.parseInt(rollNo.getText().toString());  
        db.delete(roll);  
  
        showDataInListView1();//after delete data chang in list
```



```

    }
    });

    btnsearch.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            String n = name.getText().toString();
            Cursor cursor=db.dataShowSerch(n);
            String[] key={"_id","gender","dob","time","city","course","hobby"};
            int[] id={R.id.tv1,R.id.tv2,R.id.tv3,R.id.tv4,R.id.tv5,R.id.tv6,R.id.tv7};
            SimpleCursorAdapter s=new
SimpleCursorAdapter(MainActivity.this,R.layout.simple_layout,cursor,key,id,0);
            listView.setAdapter(s);

        }
    });

}
private void showDataInListView1() {
    Cursor cursor=db.dataShow();
    String[] key={"_id","name","gender","dob","time","city","course","hobby"};
    int[] id={R.id.tv1,R.id.tv2,R.id.tv3,R.id.tv4,R.id.tv5,R.id.tv6,R.id.tv7,R.id.tv8};
    SimpleCursorAdapter s=new
SimpleCursorAdapter(MainActivity.this,R.layout.simple_layout,cursor,key,id,0);
    listView.setAdapter(s);
}

private void timedata()
{
    final Calendar calendar = Calendar.getInstance();
    int hour = calendar.get(Calendar.HOUR_OF_DAY);
    int minute = calendar.get(Calendar.MINUTE);

    TimePickerDialog timePickerDialog=new TimePickerDialog(this, new
TimePickerDialog.OnTimeSetListener() {
        @Override
        public void onTimeSet(TimePicker view, int hourOfDay, int minute) {
            selectedTime = hourOfDay + ":" + minute;
            timetv.setText("Selected Time: " + selectedTime);
        }
    }, hour, minute, true);

    timePickerDialog.show();
}

```

```

private void selectcourse() {
    final String[] course = {"MCA", "PGDCA", "BBA"};

    AlertDialog.Builder builder = new AlertDialog.Builder(this);
    builder.setTitle("Choose One Option")
        .setSingleChoiceItems(course, -1, (dialog, which) -> selectedCourse = course[which])
        .setPositiveButton("OK", (dialog, which) -> {
            coursetv.setText("Selected: " + selectedCourse);
            dialog.dismiss();
        })
        .setNegativeButton("Cancel", (dialog, which) -> dialog.dismiss());

    AlertDialog dialog = builder.create();
    dialog.show();
}
}

```

simple_layout

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="horizontal">

    <TextView
        android:id="@+id/tv1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Roll"
        android:textSize="20dp" />

    <TextView
        android:id="@+id/tv2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Name"
        android:textSize="20dp" />

    <TextView
        android:id="@+id/tv3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"

```

```
    android:layout_weight="1"
    android:text="Gender"
    android:textSize="20dp" />
```

```
<TextView
    android:id="@+id/tv4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:text="date"
    android:textSize="20dp" />
```

```
<TextView
    android:id="@+id/tv5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:text="time"
    android:textSize="20dp" />
```

```
<TextView
    android:id="@+id/tv6"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:text="city"
    android:textSize="20dp" />
```

```
<TextView
    android:id="@+id/tv7"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:text="course"
    android:textSize="20dp" />
```

```
<TextView
    android:id="@+id/tv8"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_weight="1"
    android:text="hobby"
    android:textSize="20dp" />
```

```
</LinearLayout>
```

Dbhelper

```
package com.example.crud;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

import androidx.annotation.Nullable;

class DbHelper extends SQLiteOpenHelper {
    public DbHelper(@Nullable Context context) {
        super(context, "Demo.db", null, 1);
    }
    @Override
    public void onCreate(SQLiteDatabase sqLiteDatabase) {
        String q="create table stud(rno integer primary key,name text,gender text,dob date,time
text,city text,course text,hobby text)";
        sqLiteDatabase.execSQL(q);
    }
    @Override
    public void onUpgrade(SQLiteDatabase sqLiteDatabase, int i, int i1) {
        sqLiteDatabase.execSQL("drop table if exists stud");
        onCreate(sqLiteDatabase);
    }
    public void insert(int roll, String name, String gender, String date,String time,String city,String
course,String hobby){
        SQLiteDatabase db=this.getWritableDatabase();
        ContentValues cv=new ContentValues();
        cv.put("rno",roll);
        cv.put("name",name);
        cv.put("gender",gender);
        cv.put("dob", date);
        cv.put("time", time);
        cv.put("city", city);
        cv.put("course", course);
        cv.put("hobby", hobby);
        db.insert("stud",null,cv);
    }
    public void update(int roll, String name, String gender, String date, String time, String city, String
course, String hobby){
        SQLiteDatabase db=this.getWritableDatabase();
        ContentValues cv=new ContentValues();
        cv.put("name",name);
        cv.put("gender",gender);
        cv.put("dob", date);
```

```

        cv.put("time", time);
        cv.put("city", city);
        cv.put("course", course);
        cv.put("hobby", hobby);
        db.update("stud",cv,"rno=?",new String[]{String.valueOf(roll)});
    }
    public void delete(int roll){
        SQLiteDatabase db=this.getWritableDatabase();
        db.delete("stud","rno=?",new String[]{String.valueOf(roll)});
    }
    public Cursor dataShow(){
        SQLiteDatabase db=this.getWritableDatabase();
        return db.rawQuery("select rno AS _id, name, gender, dob, time, city, course, hobby from stud",null);
    }

    public Cursor dataShowSerch(String name){
        SQLiteDatabase db=this.getWritableDatabase();
        return db.rawQuery("select rno AS _id, gender, dob, time, city, course, hobby from stud where name=?",new String[]{name});
    }
}

```